Boundary Information Geometry for S-box Side-Channel Leakage: Walsh Covariance, Active Fisher Structure, and CPA Belief Dynamics

Abstract

Side-channel attacks on lightweight block ciphers often reduce locally to distinguishing key hypotheses through leakage generated by bijective S-boxes. This paper develops an information-geometric model for these finite key-induced distributions while separating three security-relevant spaces: the singular distributions induced by exact key hypotheses, the regular full-support exponential family used for Fisher geometry, and the adversary's posterior belief simplex. For an s-bit bijective S-box S and key hypothesis k, the exact distribution $P_k(a,b) = 2^{-s}\mathbf{1}[b = S(a \oplus k)]$ is supported on one graph inside $\{0,1\}^{2s}$ and therefore lies on the boundary of the probability simplex. Walsh characters give the key-translation rule $\widehat{P}_k(\alpha,\beta) = (-1)^{\alpha\cdot k}\widehat{P}_0(\alpha,\beta)$. Using this rule as a structural input, we derive closed-form boundary covariance matrices and prove that, for every bijective S-box, their active subspace has dimension $2^s - 1$ and active eigenvalues equal 2^s in the unnormalised Walsh basis. We then interpret correlation power analysis under Gaussian leakage as Bayesian dynamics on the key-belief simplex. The Hamming-weight variance $s/(4\sigma^2)$ is a bijectivity-driven baseline, whereas pairwise likelihood gaps determine key-hypothesis discrimination and worst-case posterior-odds growth. Reproducible computations and Monte Carlo CPA simulations for PRESENT, GIFT, and SKINNY show how S-boxes with

the same Walsh-amplitude distribution can have distinct hard and easy key differences under Hamming-weight leakage while sharing the boundary covariance invariants forced by bijectivity.

Keywords: cybersecurity, side-channel analysis, correlation power analysis, S-box, information geometry, Fisher–Rao metric, Walsh–Hadamard transform, Bayesian inference

1 Introduction

1.1 Motivation

Side-channel attacks use implementation leakage in addition to the abstract cipher description. In many attacks on lightweight block ciphers, the local statistical object is a keyed S-box computation: an adversary observes or controls an input nibble or byte, measures a noisy leakage trace, and compares key hypotheses through predicted intermediate values. Correlation power analysis (CPA) is the standard example: each trace updates the adversary's belief about the key using a leakage model such as Hamming weight.

The exact distribution induced by a keyed bijective S-box is not a smooth interior point of a probability simplex. For an s-bit bijective S-box $S : \{0,1\}^s \to \{0,1\}^s$ and key hypothesis $k \in \{0,1\}^s$, the pair (a,b) with uniform a and $b = S(a \oplus k)$ is supported on only 2^s points of the 2^{2s}-point input-output space. Hence P_k is a singular boundary distribution. This matters because ordinary Fisher–Rao tensors, geodesics, and dual-flat coordinates are defined on full-support statistical models, not directly at support-deficient endpoints.

This paper develops a boundary information-geometric framework for this side-channel setting. The central discipline is to keep separate three objects that are often conflated in informal geometric descriptions:

(i) the regular full-support Walsh exponential family, where ordinary Fisher geometry is valid;

(ii) the finite boundary orbit of exact key-induced S-box distributions;

(iii) the adversary's posterior belief simplex over key hypotheses, where CPA updates take place.

This separation makes it possible to use information geometry in a controlled way: boundary distributions are represented through covariance limits, whereas Bayesian key recovery remains a regular trajectory on the belief simplex as long as the prior has full support and the leakage likelihood is positive.

For cybersecurity and hardware-security evaluation, the practical output of this separation is a diagnostic workflow for identifying which quantities are unavoidable consequences of bijectivity, which quantities are artefacts of the chosen Walsh coordinates, and which quantities actually affect CPA discrimination. In particular, for Hamming-weight leakage the scalar variance is identical for all bijective S-boxes of the same width, but the pairwise gaps between key hypotheses need not be identical. Those gaps determine the expected posterior-odds growth against wrong keys and therefore identify the key differences that are locally hardest or easiest for a CPA adversary to reject.

1.2 Relation to prior work

The information-geometric background is the Fisher–Rao and dual-affine geometry of finite statistical models and regular exponential families [1–3]. The closest coding-theoretic precedent is the work of Ikeda, Tanaka, and Amari on LDPC and turbo codes [4]: parity-check statistics define a regular information-geometric model and

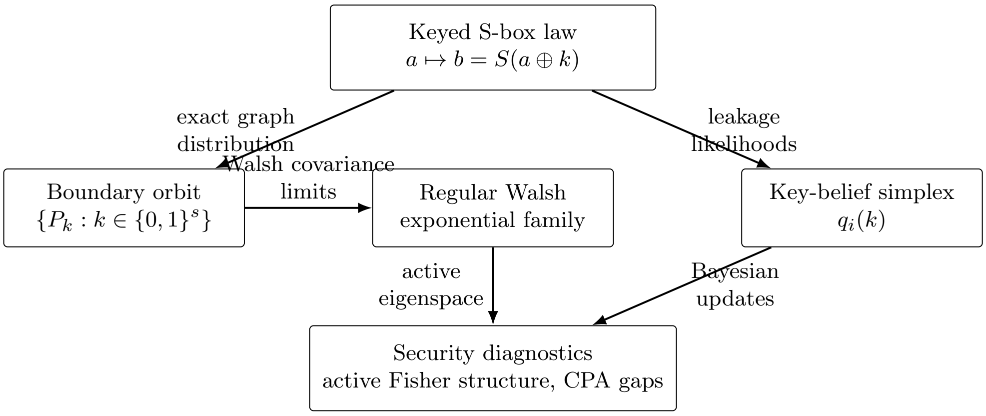

Fig. 1 Boundary-to-belief framework. The exact keyed S-box laws lie on a singular boundary orbit, the Fisher calculations are made in a regular Walsh exponential family or in boundary covariance limits, and CPA updates evolve on the separate key-belief simplex.

decoding trajectories are interpreted through projections. Here, the sufficient statistics are Walsh–Hadamard characters on S-box input-output pairs, and the inference trajectory is the CPA posterior over key hypotheses.

On the cryptographic side, the work is related to classical Walsh/LAT and DDT analysis of vectorial Boolean functions and S-boxes, including Nyberg's differential uniformity [5]. Four-bit S-boxes and optimal 4-bit permutations have been classified in the Boolean-function literature, notably by Leander and Poschmann [6]. Recent geometric viewpoints in symmetric cryptanalysis also include geometric formulations of linear and differential cryptanalysis [7, 8]. The present paper builds on this background by asking which information-geometric objects are associated with exact keyed S-box distributions and how those objects connect to side-channel belief updates. This positioning uses existing 4-bit S-box classifications as prior cryptographic knowledge and focuses on the local leakage distributions used in CPA.

On the side-channel side, the leakage model and posterior update are consistent with differential and correlation power analysis [9, 10] and with information-theoretic analyses of key recovery attacks [11, 12]. The present framework gives a formal model

Table 1 Relation to existing metrics.

Metric or framework	Standard role	Role in this paper
Walsh/LAT spectrum	Measures linear correlations of vectorial Boolean functions and supports nonlinearity analysis.	Provides coordinates for the key-induced boundary orbit and for closed-form covariance diagnostics.
DDT and differential uniformity	Measures differential propagation and classical S-box resistance to differential attacks [5].	Used as part of the surrounding S-box context for separating classical cryptographic criteria from leakage-geometry diagnostics.
4-bit S-box classification	Classifies small permutations up to established equivalences [6].	Used as prior cryptographic knowledge for interpreting the local leakage geometry.
CPA and DPA likelihood models	Explain key recovery from noisy implementation leakage [9, 10].	Interpreted as Bayesian dynamics on the key-belief simplex with pairwise likelihood gaps.
Mutual information and key-rank measures	Quantify leakage and attack success at the experiment or trace-distribution level [11, 12].	Complemented by local structural diagnostics for keyed S-box laws and posterior-odds growth.
Information geometry of codes	Studies decoding and projections in regular statistical models [4].	Adapted to singular S-box graph distributions by separating boundary covariance, regular interiors, and belief-simplex updates.

for separating baseline invariants forced by bijectivity from pairwise leakage gaps that control key-hypothesis discrimination.

Table 1 summarizes the intended position of the framework relative to standard S-box and side-channel criteria.

1.3 Setting and notation

Throughout, $s \geq 1$ denotes the S-box width in bits, $S : \{0,1\}^s \to \{0,1\}^s$ is a fixed bijection, $k \in \{0,1\}^s$ is a key hypothesis, and $a \in \{0,1\}^s$ is a uniformly distributed plaintext nibble or byte. All additions inside bit strings are over $\mathbb{F}_2$. The inner product is $\alpha \cdot a = \bigoplus_{i=1}^{s} \alpha_i a_i$. For $(\alpha, \beta) \in \{0,1\}^s \times \{0,1\}^s$, the Walsh character is

$$\chi_{\alpha\beta}(a, b) = (-1)^{\alpha \cdot a \oplus \beta \cdot b}.$$

We write $N = 2^s$ and $\mathcal{X} = \{0,1\}^s \times \{0,1\}^s$.

1.4 Contributions

C1. Boundary model for key-induced S-box distributions. We model the exact keyed S-box law

$$P_k(a, b) = 2^{-s}\mathbf{1}[b = S(a \oplus k)]$$

as a boundary distribution on the input-output simplex. The Walsh translation rule

$$\widehat{P}_k(\alpha, \beta) = (-1)^{\alpha \cdot k}\widehat{P}_0(\alpha, \beta)$$

is used as a coordinate identity for the finite key orbit and as the algebraic input for the covariance and leakage diagnostics.

C2. Closed-form boundary covariance and active Fisher structure. We derive the boundary covariance form

$$\mathcal{I}^{\partial}_{ij}(k) = \mathrm{Cov}_{P_k}(T_i, T_j)$$

in Walsh coordinates and prove that, for every bijection $S : \{0, 1\}^s \to \{0, 1\}^s$, this matrix has rank $2^s - 1$ and active eigenvalues all equal to 2^s in the unnormalised Walsh basis. Thus the active boundary geometry is a structural property of bijective S-box graphs shared by the three illustrative 4-bit examples.

C3. Regular interior and belief-simplex geometry. We explicitly separate the regular Walsh exponential family, where the usual Amari dual-flat structure and Pythagorean identities apply, from the singular boundary orbit. The adversary's key belief is treated as a separate full- support simplex, equipped with its own Fisher–Rao metric.

C4. CPA as Bayesian belief dynamics with pairwise leakage gaps. Under Gaussian leakage, each CPA trace is represented as a Bayesian update on the key-belief simplex. The Hamming-weight variance $s/(4\sigma^2)$ is identified as a baseline forced by bijectivity and uniform inputs. The quantities that distinguish key hypotheses are the pairwise separations

$$\Delta(k, k^*) = \frac{1}{2^s} \sum_a \big(f(S(a \oplus k^*)) - f(S(a \oplus k))\big)^2,$$

which determine expected log-Bayes gains.

C5. **Reproducible diagnostics on lightweight S-boxes.** For PRESENT, GIFT, and SKINNY, we compute Walsh spectra, boundary covariance matrices, local spectral separation proxies, pairwise Hamming-weight leakage gaps, Monte Carlo CPA success curves, and the projected active-eigenspace curvature diagnostic. The examples are used to separate invariants forced by bijectivity or by the shared Walsh amplitude distribution from quantities that can vary across S-boxes. The pairwise-gap tables and CPA simulations identify the hard key differences and trace-dependent key-rank behavior under the assumed leakage model.

1.5 Model coverage

The paper focuses on isolated bijective S-boxes under uniform inputs and on an additive Gaussian leakage model for CPA belief updates. This coverage matches the local leakage setting in which S-box output predictions drive non-profiled CPA. Extensions to complete multi-round ciphers, masking, higher-order leakage, non-Gaussian leakage, profiled attacks, and implementation-specific trace effects can be built by adding the corresponding leakage and cipher propagation models to the boundary-distribution and belief-update framework.

2 Background

2.1 S-boxes and the Walsh–Hadamard transform

An *s-bit S-box* is a bijection $S : \{0,1\}^s \to \{0,1\}^s$. The i-th output bit is a Boolean function $S_i : \{0,1\}^s \to \{0,1\}$.

Definition 2.1 (Walsh–Hadamard Transform) The Walsh–Hadamard transform (WHT) of S at $(\alpha, \beta) \in \{0,1\}^s \times \{0,1\}^s$ is:

$$\widehat{S}(\alpha, \beta) = \sum_{a \in \{0,1\}^s} (-1)^{\alpha \cdot a \,\oplus\, \beta \cdot S(a)} \;\in\; \{-2^s, \ldots, +2^s\}, \qquad 2 \mid \widehat{S}(\alpha, \beta). \tag{1}$$

By convention $\widehat{S}(0,0) = 2^s$.

The WHT encodes the two fundamental cryptographic properties of S:

- **Nonlinearity**: $\mathrm{NL}(S) = 2^{s-1} - \frac{1}{2} \max_{(\alpha,\beta) \neq (0,0)} |\widehat{S}(\alpha, \beta)|$.
- **Differential uniformity**: $\delta(S) = \max_{\Delta_{\mathrm{in}} \neq 0,\, \Delta_{\mathrm{out}}} \big|\{a : S(a) \oplus S(a \oplus \Delta_{\mathrm{in}}) = \Delta_{\mathrm{out}}\}\big|$.

The *Linear Approximation Table* (LAT) entry is $\mathrm{LAT}_S(\alpha, \beta) = \widehat{S}(\alpha, \beta)/2$. For any S-box, Parseval's identity gives:

$$\sum_{(\alpha,\beta) \in \{0,1\}^s \times \{0,1\}^s} \widehat{S}(\alpha, \beta)^2 = 2^{2s} \cdot 2^s = 2^{3s}. \tag{2}$$

Example 2.1 (PRESENT S-box) The PRESENT-80 S-box ($s = 4$) is the hexadecimal table $S =$ `[C,5,6,B,9,0,A,D,3,E,F,8,4,7,1,2]`. Its nonlinearity is $\mathrm{NL}(S) = 4$ and differential uniformity $\delta(S) = 4$.

2.1.1 Remark on 4-bit S-box classification

The 4-bit case is small enough that strong classification results are available. In particular, optimal 4-bit S-boxes have been studied up to the standard equivalence relations used for vectorial Boolean functions [6]. The numerical examples in Section 7 use this classification background to provide reproducible side-channel diagnostics for widely used lightweight S-boxes and illustrate which information-geometric quantities are forced by bijectivity or by common Walsh spectra.

2.2 Information geometry: essentials

We collect the minimal background from Amari and Nagaoka [1] and Ay et al. [2] required in the sequel.

2.2.1 Statistical manifolds and the Fisher metric

Let $\mathcal{X}$ be a finite set with $|\mathcal{X}| = N$. The *probability simplex* is $\Delta_{N-1} = \{p \in \mathbb{R}^N : p_x \geq 0,\ \sum_x p_x = 1\}$. Its interior $\mathring{\Delta}_{N-1}$ is an $(N-1)$-dimensional manifold.

Definition 2.2 (Fisher–Rao Metric) The *Fisher–Rao metric* on $\mathring{\Delta}_{N-1}$ at p is the positive definite bilinear form

$$g_p(u, v) = \sum_{x \in \mathcal{X}} \frac{u_x\, v_x}{p_x}, \qquad u, v \in T_p\mathring{\Delta}_{N-1} = \Big\{ w \in \mathbb{R}^N : \sum_x w_x = 0 \Big\}. \tag{3}$$

By Chentsov's theorem [3], g_F is the unique Riemannian metric on $\mathring{\Delta}_{N-1}$ (up to a positive constant) that is invariant under sufficient statistics, making it the canonical statistical distance.

2.2.2 Exponential families as flat submanifolds

Definition 2.3 (Exponential Family) Given functions $T_1, \ldots, T_d : \mathcal{X} \to \mathbb{R}$ (the *sufficient statistics*), the *exponential family* $\mathcal{E}$ is:

$$p(x;\theta) = \exp\Big(\sum_{i=1}^{d} \theta_i T_i(x) - \psi(\theta)\Big), \quad \theta \in \Theta \subseteq \mathbb{R}^d, \tag{4}$$

where $\psi(\theta) = \log \sum_x \exp(\sum_i \theta_i T_i(x))$ is the log-partition function.

The natural parameters θ_i and the expectation parameters $\eta_i = \mathbb{E}_\theta[T_i(X)]$ are related by $\eta = \nabla\psi(\theta)$ (the bijection is guaranteed when the family is minimal and regular).

Theorem 2.1 (Dually Flat Structure, [1]) *Every regular minimal exponential family $\mathcal{E}$ is a dually flat submanifold of $\mathring{\Delta}_{N-1}$: the e-connection (exponential connection) is flat in natural parameters θ, and the m-connection (mixture connection) is flat in expectation parameters η. The generalised Pythagorean theorem holds: if $P_A, P_B, P_C \in \mathcal{E}$ and the e-geodesic from A to B is g_F-orthogonal to the m-geodesic from B to C, then*

$$D_{\mathrm{KL}}(P_A \| P_C) = D_{\mathrm{KL}}(P_A \| P_B) + D_{\mathrm{KL}}(P_B \| P_C). \tag{5}$$

2.2.3 Mixture projection and Bayesian inference

Definition 2.4 (m-Projection) Given $p \in \mathring{\Delta}_{N-1}$ and a closed m-flat submanifold $\mathcal{M} \subseteq \mathring{\Delta}_{N-1}$, the *$m$-projection* of p onto $\mathcal{M}$ is:

$$\Pi^{(m)}_{\mathcal{M}}(p) = \arg\min_{q \in \mathcal{M}} D_{\mathrm{KL}}(q \| p). \tag{6}$$

The following classical result (see [1], Chapter 3) is central to Section 5:

Proposition 2.1 (Bayesian Update as m-Projection) *Let $p(x, y) = p(x)\, p(y|x)$ be a joint distribution on $\mathcal{X} \times \mathcal{Y}$. After observing $y = y_0$, the posterior $p(x|y_0)$ is the m-projection of* *the joint p onto the submanifold $\mathcal{M}_{y_0} = \{q \in \Delta : q(y) = 0$ for $y \neq y_0\}$, restricted to the x-marginal.*

2.2.4 The LDPC analogy: Ikeda-Tanaka-Amari (2004)

Ikeda, Tanaka, and Amari [4] develop a complete information-geometric framework for LDPC and turbo codes. They model the joint distribution over a binary codeword $x \in \{-1, +1\}^N$ as:

$$p(x; \theta, v) = \exp\Big(\theta \cdot x + \sum_r v_r \, c_r(x) - \psi\Big), \tag{7}$$

where $c_r(x) = \prod_{i \in r} x_i$ are parity check functions and v_r the constraint weights. They show: (i) the e/m dual structure is exact; (ii) Belief Propagation decoding corresponds to iterated e/m-projections; (iii) curvature of the statistical manifold controls decoding convergence.

In the present paper, the parity functions $c_r(x)$ are replaced by the Walsh–Hadamard characters $\chi_{\alpha\beta}(a, b)$ of the S-box, and LDPC decoding is replaced by CPA key recovery. Table 2 summarises the correspondence.

Table 2 Structural analogy between the LDPC framework of Ikeda-Tanaka-Amari (2004) and the S-box framework of this paper.

Ikeda-Tanaka-Amari (LDPC)	This paper (S-box)
Codeword $x \in \{-1, +1\}^N$	Input-output pair $(a, b) \in \{0, 1\}^{2s}$
Parity check $c_r(x) = \prod_{i \in r} x_i$	Walsh character $\chi_{\alpha\beta}(a, b) = (-1)^{\alpha \cdot a \oplus \beta \cdot b}$
Channel log-likelihood θ_i	Key log-likelihood ratio $\log(q_k / q_{k'})$
Parity constraint weight v_r	Walsh spectrum value $\widehat{S}(\alpha, \beta)$
Belief Propagation decoding	CPA key recovery
Decoding convergence rate	Per-trace signal information $I_S(\sigma)$

3 The S-box Statistical Model

3.1 Key-induced boundary distributions

Fix a bijective S-box $S : \{0,1\}^s \to \{0,1\}^s$ and let A be uniformly distributed over $\{0,1\}^s$. Under key hypothesis $k \in \{0,1\}^s$, the observed input-output pair is $(A, S(A \oplus k))$. This induces the following distribution on $\mathcal{X} = \{0,1\}^s \times \{0,1\}^s$.

Definition 3.1 (Key-induced distribution) For $k \in \{0,1\}^s$, define

$$P_k(a,b) = 2^{-s}\mathbf{1}[b = S(a \oplus k)]. \tag{8}$$

The support of P_k is the graph $G_k = \{(a, S(a \oplus k)) : a \in \{0,1\}^s\}$.

Each P_k is a boundary point of the simplex $\Delta_{2^{2s}-1}$, because it assigns zero mass to $2^{2s} - 2^s$ outcomes. This support deficiency is central: ordinary Fisher–Rao tensors of the full simplex are defined on the interior, not directly at P_k. We therefore treat P_k as a singular endpoint of an interior regularization when differential-geometric operations are needed.

3.2 Walsh representation and key translation

Definition 3.2 (Walsh coefficient of P_k) For $(\alpha, \beta) \in \{0,1\}^s \times \{0,1\}^s$, set

$$\widehat{P}_k(\alpha,\beta) = \sum_{(a,b)\in\mathcal{X}} \chi_{\alpha\beta}(a,b) P_k(a,b) = 2^{-s} \sum_{a\in\{0,1\}^s} (-1)^{\alpha\cdot a \oplus \beta\cdot S(a\oplus k)}. \tag{9}$$

Lemma 3.1 (Walsh key-translation rule) *For every $k \in \{0,1\}^s$ and every $(\alpha, \beta) \in \{0,1\}^s \times \{0,1\}^s$,*

$$\widehat{P}_k(\alpha,\beta) = (-1)^{\alpha\cdot k}\widehat{P}_0(\alpha,\beta) = 2^{-s}(-1)^{\alpha\cdot k}\widehat{S}(\alpha,\beta). \tag{10}$$

Proof Using $a' = a \oplus k$,

$$\widehat{P}_k(\alpha,\beta) = 2^{-s} \sum_{a'} (-1)^{\alpha\cdot(a'\oplus k)\oplus\beta\cdot S(a')} = 2^{-s}(-1)^{\alpha\cdot k} \sum_{a'} (-1)^{\alpha\cdot a'\oplus\beta\cdot S(a')},$$

which is exactly the claimed identity. □

Remark 3.1 (Walsh orbit) The key does not change the amplitude $|\widehat{S}(\alpha,\beta)|$; it only changes the sign according to the character $\alpha \mapsto (-1)^{\alpha\cdot k}$. Thus $\{P_k : k \in \{0,1\}^s\}$ is best described as a finite Walsh-phase orbit, not as a smooth manifold. The rule is a structural translation identity; the information-geometric content in the sequel comes from the covariance limit, active subspace, and belief-update geometry built on top of this identity.

3.3 The regular Walsh exponential family

Definition 3.3 (Walsh exponential family) Let $T_{\alpha\beta} = \chi_{\alpha\beta}$ for $(\alpha,\beta) \neq (0,0)$. The Walsh exponential family on $\mathcal{X}$ is

$$p_\theta(a,b) = \exp\left(\sum_{(\alpha,\beta)\neq(0,0)} \theta_{\alpha\beta} T_{\alpha\beta}(a,b) - \psi(\theta) \right), \qquad \theta \in \mathbb{R}^{2^{2s}-1}. \tag{11}$$

Because the nonconstant Walsh characters together with the constant character form an orthogonal basis of all real functions on $\mathcal{X}$, this is simply the full regular exponential family over $\mathcal{X}$ written in Walsh coordinates. Its image is the interior of the probability simplex. The singular distributions P_k belong to the closure, not to the regular parameter space.

Proposition 3.1 (Boundary expectation coordinates) *For each k and $(\alpha,\beta) \neq (0,0)$,*

$$\eta^{(k)}_{\alpha\beta} := \mathbb{E}_{P_k}[T_{\alpha\beta}] = (-1)^{\alpha\cdot k} 2^{-s} \widehat{S}(\alpha,\beta). \tag{12}$$

The corresponding natural parameters are not finite; they are obtained only as limits of full-support approximations approaching P_k.

Proof The first equality is the definition of expectation coordinates and the second is Lemma 3.1. Since P_k has zeros on $\mathcal{X}$, no finite vector θ in (11) can represent it. □

3.4 Interior regularization and boundary covariance form

Let U denote the uniform distribution on $\mathcal{X}$. For $\varepsilon \in (0,1)$ define

$$P_k^{\varepsilon} = (1-\varepsilon)P_k + \varepsilon U. \tag{13}$$

Then P_k^{ε} has full support and belongs to the regular simplex interior. Fisher–Rao calculations are unambiguous at P_k^{ε}. The finite covariance limit as $\varepsilon \downarrow 0$ is the following.

Definition 3.4 (Boundary Fisher covariance form) The boundary covariance form associated with key k is

$$\mathcal{I}^{\partial}_{ij}(k) = \mathrm{Cov}_{P_k}(T_i, T_j), \qquad i,j \in \{0,1\}^s \times \{0,1\}^s \setminus \{(0,0)\}. \tag{14}$$

It is positive semidefinite and generally singular. We do not call it a non-degenerate Fisher metric at P_k.

Proposition 3.2 (Closed form of the boundary covariance) *For $i = (\alpha, \beta)$ and $j = (\alpha', \beta')$,*

$$\begin{aligned}\mathcal{I}^{\partial}_{ij}(k) &= \mathrm{Cov}_{P_k}(T_{\alpha\beta}, T_{\alpha'\beta'}) \\ &= (-1)^{(\alpha\oplus\alpha')\cdot k}\left(2^{-s}\widehat{S}(\alpha\oplus\alpha', \beta\oplus\beta') - 2^{-2s}\widehat{S}(\alpha,\beta)\widehat{S}(\alpha',\beta')\right).\end{aligned} \tag{15}$$

In particular,

$$\mathcal{I}^{\partial}_{(\alpha\beta),(\alpha\beta)}(k) = 1 - \left(2^{-s}\widehat{S}(\alpha,\beta)\right)^2. \tag{16}$$

Proof Since $T_i T_j = T_{i\oplus j}$,

$$\mathbb{E}_{P_k}[T_i T_j] = \widehat{P}_k(\alpha\oplus\alpha', \beta\oplus\beta') = (-1)^{(\alpha\oplus\alpha')\cdot k} 2^{-s}\widehat{S}(\alpha\oplus\alpha', \beta\oplus\beta').$$

Subtracting $\mathbb{E}_{P_k}[T_i]\mathbb{E}_{P_k}[T_j]$ and applying Lemma 3.1 gives (15). The diagonal follows from $T_i^2 = 1$. □

Lemma 3.2 (Key covariance isometry) *Let D_k be the diagonal sign matrix $(D_k)_{(\alpha\beta),(\alpha\beta)} = (-1)^{\alpha\cdot k}$. Then*

$$\mathcal{I}^{\partial}(k) = D_k \mathcal{I}^{\partial}(0) D_k. \tag{17}$$

Thus every spectral invariant of the boundary covariance form is independent of the true key.

Proof Equation (15) factors exactly into the sign contribution $(-1)^{\alpha\cdot k}(-1)^{\alpha'\cdot k}$ times the $k = 0$ covariance entry. □

Remark 3.2 (What is metric and what is not) For $\varepsilon > 0$, the Fisher–Rao metric at P_k^{ε} is an ordinary Riemannian metric. At $\varepsilon = 0$, only the finite covariance form (14) is used. Distances or curvatures involving the exact P_k are therefore either (i) computed on the belief simplex, (ii) computed for P_k^{ε} and then studied as limits, or (iii) explicitly described as first-order/local covariance approximations.

3.5 First-order spectral separation of key hypotheses

The exact distributions P_k and $P_{k'}$ have disjoint supports when $k \neq k'$, so their KL divergence is infinite. Nevertheless, the Walsh coordinate displacement has a simple closed form:

$$\eta_{\alpha\beta}^{(k')} - \eta_{\alpha\beta}^{(k)} = \left((-1)^{\alpha\cdot k'} - (-1)^{\alpha\cdot k}\right) 2^{-s}\widehat{S}(\alpha,\beta). \tag{18}$$

Consequently, only Walsh modes satisfying $\alpha\cdot(k\oplus k') = 1$ contribute to the spectral separation. A diagonal local approximation based on the boundary covariance form is

$$d_{\mathrm{loc}}^2(k,k') = \sum_{\substack{(\alpha,\beta)\neq(0,0):\\ \alpha\cdot(k\oplus k')=1}} \frac{4\,2^{-2s}\widehat{S}(\alpha,\beta)^2}{1 - 2^{-2s}\widehat{S}(\alpha,\beta)^2}, \tag{19}$$

provided the denominator is nonzero. This quantity is a local spectral proxy, not the exact Fisher–Rao geodesic distance between the singular distributions.

4 Dual Structure, Boundary Limits, and Belief Geometry

4.1 Regular dual flatness of the interior family

Theorem 4.1 (Regular dual flatness) *The Walsh exponential family* (11) *is a regular minimal exponential family on $\mathcal{X}$. Its interior carries the standard dually flat information-geometric structure: natural coordinates θ are e-affine, expectation coordinates η are m-affine, the log-partition function ψ is the e-potential, and its Legendre dual is $\varphi(\eta) = -H(p_\eta)$.*

Proof The natural parameter space is the open set $\mathbb{R}^{2^{2s}-1}$. The sufficient statistics are bounded, so the partition function is finite and smooth for all θ. Minimality follows from the orthogonality and linear independence of the nonconstant Walsh characters together with the constant character. The standard theorem for regular minimal exponential families then gives dual flatness and the Legendre correspondence [1, 2]. □

Remark 4.1 (Boundary discipline) Theorem 4.1 applies to full-support distributions only. The exact key distributions P_k lie in the closure. Statements about geodesics, Christoffel symbols, Bregman identities, or curvature at P_k are therefore meaningful only after regularization or after being rephrased as finite boundary covariance statements.

4.2 KL divergence between exact key distributions

Proposition 4.1 (Support separation) *If S is bijective and $k \neq k'$, then*

$$\operatorname{supp}(P_k) \cap \operatorname{supp}(P_{k'}) = \varnothing, \qquad D_{\mathrm{KL}}(P_k \| P_{k'}) = +\infty. \tag{20}$$

Proof If (a,b) belongs to both supports, then $S(a \oplus k) = b = S(a \oplus k')$. Since S is injective, $a \oplus k = a \oplus k'$, hence $k = k'$, a contradiction. Thus $P_{k'}$ assigns zero probability to every point in the support of P_k when $k \neq k'$, which implies infinite KL divergence. □

Remark 4.2 (No finite Bregman identity at singular endpoints) The usual equality between KL divergence and the Bregman divergence of the dual potential is an interior statement. For P_k and $P_{k'}$ with disjoint supports, one may study $D_{\mathrm{KL}}(P_k^{\varepsilon} \| P_{k'}^{\varepsilon})$ and then let $\varepsilon \downarrow 0$, but the limiting divergence is infinite. We therefore do not use finite Pythagorean decompositions directly between distinct exact key distributions.

4.3 Mixtures induced by key beliefs

Let $Q = \Delta^{\circ}_{2^s-1}$ be the interior of the simplex over keys. A belief $q \in Q$ induces a distribution on input-output pairs

$$\bar{P}_q(a,b) = \sum_{k \in \{0,1\}^s} q(k) P_k(a,b) = 2^{-s} \sum_{k \in \{0,1\}^s} q(k) \mathbf{1}[b = S(a \oplus k)]. \tag{21}$$

If q has full support, then $\bar{P}_q$ has full support on $\mathcal{X}$.

Proposition 4.2 (Walsh representation of belief mixtures) *Let*

$$\widehat{q}(\alpha) = \sum_{k \in \{0,1\}^s} q(k)(-1)^{\alpha \cdot k}.$$

Then

$$\bar{P}_q(a,b) = 2^{-2s} \sum_{\alpha,\beta \in \{0,1\}^s} \widehat{q}(\alpha) \widehat{S}(\alpha,\beta)(-1)^{\alpha \cdot a \oplus \beta \cdot b}. \tag{22}$$

In particular, the uniform belief induces the uniform distribution on $\mathcal{X}$.

Proof Apply Walsh inversion to the indicator $\mathbf{1}[b = S(a \oplus k)]$ and sum over k with weights $q(k)$. For uniform q, $\widehat{q}(\alpha) = \mathbf{1}[\alpha = 0]$, and $\widehat{S}(0,0) = 2^s$ while $\widehat{S}(0,\beta) = 0$ for $\beta \neq 0$ because S is bijective. □

4.4 Fisher–Rao geometry of the belief simplex

Definition 4.1 (Belief simplex metric) For $q \in Q$ and tangent vectors u, v with $\sum_k u_k = \sum_k v_k = 0$, the Fisher–Rao metric is

$$g_q^{(Q)}(u, v) = \sum_{k \in \{0,1\}^s} \frac{u_k v_k}{q(k)}. \tag{23}$$

Under the square-root map $q \mapsto (\sqrt{q(k)})_k$, this metric is the round metric on a sphere of radius 2. Therefore

$$d_F(q, q') = 2 \arccos \left(\sum_{k \in \{0,1\}^s} \sqrt{q(k) q'(k)} \right). \tag{24}$$

For distinct point-mass limits, $d_F(\delta_k, \delta_{k'}) = \pi$. This is a statement about the belief simplex, not about geodesics between the singular input-output distributions P_k and $P_{k'}$.

4.5 Pythagorean identities

The generalized Pythagorean theorem is used in the following restricted sense. For full-support distributions in the regular Walsh exponential family, the standard identity

$$D_{\mathrm{KL}}(P_A \| P_C) = D_{\mathrm{KL}}(P_A \| P_B) + D_{\mathrm{KL}}(P_B \| P_C) \tag{25}$$

holds whenever the e-geodesic from A to B is orthogonal to the m-geodesic from B to C. In the CPA setting, the relevant full-support object is the posterior belief $q_n \in Q$ as long as the prior has full support and the Gaussian likelihood is strictly positive. This is why projection statements in Section 5 are made on Q or on regularized joint distributions, not directly on disjoint-support P_k endpoints.

5 Correlation Power Analysis as Bayesian Projection Dynamics

5.1 Leakage model

We consider the standard non-profiled CPA setting. The adversary knows S, observes plaintexts $a_1, \ldots, a_n$ uniformly drawn from $\{0,1\}^s$, and obtains leakage measurements

$$W_i = f(S(a_i \oplus k^*)) + \varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}(0, \sigma^2), \tag{26}$$

where k^* is the true key and $f : \{0,1\}^s \to \mathbb{R}$ is the leakage model. In the numerical experiments we take $f = \mathrm{HW}$.

5.2 Bayesian update as a mixture projection

Theorem 5.1 (CPA posterior update as m-projection) *Let $q_{i-1} \in \Delta^{\circ}_{2^s-1}$ be the prior belief over keys before observing (a_i, W_i). The posterior*

$$q_i(k) = \frac{q_{i-1}(k)\, p(W_i \mid a_i, k)}{\sum_{h \in \{0,1\}^s} q_{i-1}(h)\, p(W_i \mid a_i, h)} \tag{27}$$

is the key marginal of the regular conditional distribution obtained from the joint prior-likelihood model. Equivalently, it can be obtained as the limit of m-projections after conditioning on shrinking measurement bins around the observed continuous value.

Proof The joint distribution before conditioning is $q_{i-1}(k)p(a_i)p(W_i \mid a_i, k)$. Since the Gaussian observation is continuous, the event $W_i = w_i$ has probability zero and conditioning is understood through a regular conditional density, or equivalently as the limit obtained by conditioning on $W_i \in [w_i - \epsilon, w_i + \epsilon]$ and letting $\epsilon \downarrow 0$. The limiting conditional law has key marginal exactly (27). $\square$

The exact Sâ€‘box distributions P_k lie on the boundary of the simplex, but the belief trajectory $q_0, q_1, \ldots, q_n$ stays in the interior of the belief simplex as long as the prior has full support and the Gaussian likelihood is positive. Hence CPA can be studied as a regular trajectory on $(Q, g_F^{(Q)})$.

5.3 Per-trace signal information

Definition 5.1 (Per-trace signal information) For a leakage model f and noise variance σ^2, define

$$I_S(f, \sigma) = \sigma^{-2} \operatorname{Var}_{a \sim \mathrm{Unif}(\{0,1\}^s)} \big[f(S(a \oplus k^*)) \big]. \tag{28}$$

When S is bijective, this quantity is independent of k^*.

Proposition 5.1 (Hamming-weight specialization) *For $f = \mathrm{HW}$ and a bijective S,*

$$I_S(\mathrm{HW}, \sigma) = \frac{s}{4\sigma^2}. \tag{29}$$

In particular, for 4â€‘bit bijective Sâ€‘boxes, $I_S(\mathrm{HW}, \sigma) = 1/\sigma^2$.

Proof Bijectivity of S together with the uniformity of $a \oplus k^*$ implies that $S(a \oplus k^*)$ is uniform over $\{0,1\}^s$. The Hamming weight of a uniform s-bit word follows a binomial distribution $\mathrm{Binomial}(s, 1/2)$, whose variance is $s/4$. □

5.4 Pairwise log-likelihood increments

For a competing key $k \neq k^*$, the perâ€‘trace squared separation is

$$\Delta(k, k^*) = \frac{1}{2^s} \sum_{a \in \{0,1\}^s} \big(f(S(a \oplus k^*)) - f(S(a \oplus k)) \big)^2. \tag{30}$$

Proposition 5.2 (Expected pairwise logâ€‘Bayes gain) *Under the Gaussian leakage model,*

$$\mathbb{E}\left[\log \frac{p(W \mid A, k^*)}{p(W \mid A, k)} \right] = \frac{\Delta(k, k^*)}{2\sigma^2}, \tag{31}$$

where the expectation is over A (uniform) and W conditioned on A and k^*.

Proof For a fixed plaintext a, the two likelihoods are Gaussian with the same variance σ^2 and means $\mu_*(a) = f(S(a \oplus k^*))$, $\mu_k(a) = f(S(a \oplus k))$. The Kullbackâ€‘Leibler divergence between two univariate Gaussians with equal variance is $(\mu_*(a) - \mu_k(a))^2/(2\sigma^2)$. Averaging over a gives the result. $\square$

The worstâ€‘case pairwise gap

$$\Gamma_S(f, \sigma) = \min_{k \neq k^*} \frac{\Delta(k, k^*)}{2\sigma^2} \tag{32}$$

provides a rigorous likelihoodâ€‘separation measure, distinct from $I_S(f, \sigma)$ in general.

5.5 Model-based sample-complexity estimate

Starting from a uniform prior, the initial key uncertainty equals $s \log 2$ nats. A firstâ€‘order information estimate then yields

$$n_{\text{est}} \approx \frac{s \log 2}{I_S(f, \sigma)}. \tag{33}$$

For 4â€‘bit Hammingâ€‘weight leakage, $n_{\text{est}} \approx 4 \log 2\, \sigma^2$. This is a heuristic leakageâ€‘rate estimate. A more rigorous bound follows from the pairwise gap $\Gamma_S(f, \sigma)$: the expected logâ€‘posterior odds against any fixed wrong key grow at rate at least $\Gamma_S(f, \sigma)$. Thus reaching an expected odds margin τ against the hardest wrong key requires roughly

$$n_{\text{gap}}(\tau) \approx \frac{\tau}{\Gamma_S(f, \sigma)} = \frac{2\sigma^2 \tau}{\min_{k \neq k^*} \Delta(k, k^*)}. \tag{34}$$

This model-based estimate links the minimum pairwise gap to a trace budget for expected posterior-odds growth. Two S-boxes with the same scalar I_S can lead to different worst-case posterior-odds growth when their minimum pairwise gaps differ.

6 Active-Eigenspace Curvature: Computation and Status

6.1 What curvature is being computed

The regular Walsh exponential family is dually flat with respect to the e- and m-connections, but its Levi–Civita curvature is not automatically zero. At exact key distributions P_k, however, the ordinary Riemannian tensor of the full-support family is not directly defined because P_k lies on the boundary. This section therefore uses the following restricted object.

Definition 6.1 (Active covariance eigenspace) Let $\mathcal{I}^{\partial}(k)$ be the boundary covariance matrix in Definition 3.4. Its active eigenspace is the span of the eigenvectors with strictly positive eigenvalues. Curvature computations below are performed after projecting the covariance and third cumulant tensors to this active eigenspace.

The resulting values are curvature diagnostics of the limiting regularized geometry obtained from the active covariance eigenspace. The finite orbit $\{P_k\}$ itself is discrete and is used here through its induced boundary covariance and cumulant tensors.

6.2 Universal rank and eigenvalues of the boundary covariance

Theorem 6.1 (Rank and active eigenvalues) *For any bijection $S : \{0,1\}^s \to \{0,1\}^s$, the boundary covariance matrix $\mathcal{I}^{\partial}(k)$ has rank $2^s - 1$. Its nonzero eigenvalues are all 2^s when the feature coordinates are the unnormalised nonconstant Walsh characters used in Definition 3.3.*

Proof Let $N = 2^s$ and index rows by $a \in \{0,1\}^s$. Let X be the $N \times (2^{2s} - 1)$ feature matrix with entries $X_{a,(\alpha,\beta)} = (-1)^{\alpha\cdot a \oplus \beta\cdot S(a\oplus k)}$ for $(\alpha, \beta) \neq (0, 0)$. The boundary covariance is $N^{-1}X^{\top}HX$, where $H = I_N - N^{-1}\mathbf{1}\mathbf{1}^{\top}$ centers over the support graph. The nonzero eigenvalues of $N^{-1}X^\top HX$ equal those of $N^{-1}HXX^\top H$. For $a, a' \in \{0,1\}^s$,

$$(XX^\top)_{a,a'} = \sum_{(\alpha,\beta)\neq(0,0)} (-1)^{\alpha\cdot(a\oplus a')\oplus\beta\cdot(S(a\oplus k)\oplus S(a'\oplus k))} = N^2\mathbf{1}[a = a'] - 1,$$

because S is injective. Hence $N^{-1}HXX^\top H = N^{-1}H(N^2 I_N - \mathbf{1}\mathbf{1}^\top)H = NH$. The matrix H has rank $N-1$ and eigenvalue 1 on the centered subspace. Therefore the active eigenvalues are all $N = 2^s$. $\square$

6.3 Third cumulants and curvature protocol

Let

$$C^\partial_{ijk}(k) = \mathbb{E}_{P_k}\left[(T_i - \eta_i)(T_j - \eta_j)(T_k - \eta_k)\right] \tag{35}$$

be the boundary third cumulant tensor. Since products of Walsh characters are again Walsh characters, every entry can be computed exactly from the WHT of S and the key-translation rule. If U is an orthonormal basis of the active eigenspace and $\lambda_a = 2^s$ are the active covariance eigenvalues, the projected tensor is

$$\widetilde{C}_{abc} = \sum_{i,j,\ell} U_{ia}U_{jb}U_{\ell c}C^\partial_{ij\ell}. \tag{36}$$

The Levi–Civita curvature diagnostic used in the experiments is then computed with the standard dually-flat formula

$$R_{abab} = \frac{1}{4}\sum_c \frac{\widetilde{C}_{aac}\widetilde{C}_{bbc} - \widetilde{C}^2_{abc}}{\lambda_c}, \qquad K_{ab} = \frac{R_{abab}}{\lambda_a\lambda_b}, \quad a < b. \tag{37}$$

Both verification scripts implement Equation (37) directly.

6.4 Empirical constant-curvature finding

Proposition 6.1 (Reproducible 4-bit curvature observation) *For the PRESENT, GIFT, and SKINNY 4-bit S-boxes, the projected active-eigenspace curvature diagnostic* (37) *gives*

$$K_{ab} = -\frac{1}{4} \tag{38}$$

for all $\binom{15}{2} = 105$ *active coordinate planes, up to numerical precision in the provided scripts.*

Verification protocol The computation is reproducible from the repository scripts. The scripts build the WHT table, construct $\mathcal{I}^{\partial}$, diagonalize its active 15-dimensional eigenspace, project the third cumulant tensor, and evaluate (37). The reported value is stable across the three tested S-boxes and an additional random 4-bit bijection in `verify_curvature.py`.

These reproducible computations suggest the following conjecture: for bijective S-boxes, the active Fisher subspace associated with the Walsh boundary covariance form has constant sectional curvature $K = -1/4$, under the curvature convention and projection procedure used in this work. A symbolic proof, or a characterization of the precise hypotheses under which the conjecture holds, is left as an open problem. □

6.5 Relation to the belief simplex

The belief simplex $Q = \Delta^{\circ}_{2^s-1}$ with Fisher–Rao metric is isometric, via the square-root embedding, to a sphere of radius 2 and hence has constant sectional curvature $+1/4$. The observed value $-1/4$ in the active boundary covariance diagnostic suggests a sign-dual phenomenon between the posterior belief geometry and the limiting keyed S-box covariance geometry, which is recorded here as part of the open geometric structure of the model.

7 Reproducible Security Diagnostics

The computations in this section connect the closed-form geometry to attack-relevant side-channel diagnostics. First, they verify the exact boundary covariance and active-eigenspace formulas on standard lightweight S-boxes. Second, they separate quantities that are structural invariants from quantities that influence CPA discrimination between key hypotheses. Third, they validate the pairwise-gap interpretation through a reproducible Monte Carlo CPA experiment.

7.1 Target S-boxes

We evaluate the 4-bit bijective S-boxes used in PRESENT, GIFT, and SKINNY [13–15]:

$$\begin{aligned} S_{\mathrm{PRESENT}} &= [C, 5, 6, B, 9, 0, A, D, 3, E, F, 8, 4, 7, 1, 2], \\ S_{\mathrm{GIFT}} &= [1, A, 4, C, 6, F, 3, 9, 2, D, B, 7, 5, 0, 8, E], \\ S_{\mathrm{SKINNY}} &= [C, 6, 9, 0, 1, A, 2, B, 3, 8, 5, D, 4, E, 7, F]. \end{aligned}$$

These examples are useful because they are widely used in lightweight cryptography and because exhaustive enumeration over all inputs, masks, and key hypotheses is possible.

7.2 Walsh spectra and boundary covariance

For each S-box we compute the full WHT table

$$\widehat{S}(\alpha, \beta) = \sum_{a \in \{0,1\}^4} (-1)^{\alpha \cdot a \oplus \beta \cdot S(a)}.$$

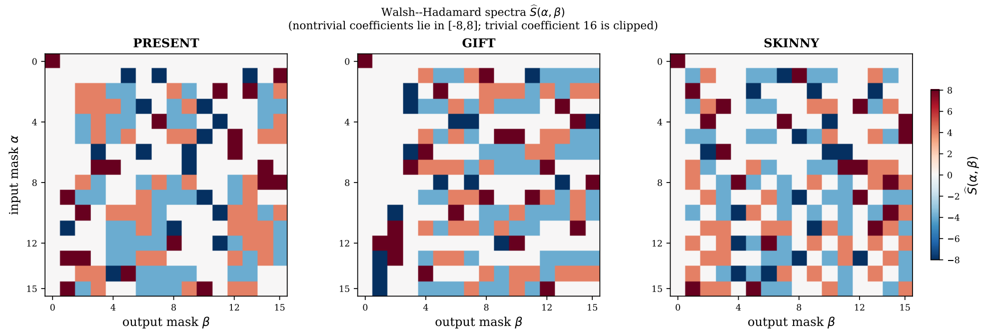

Fig. 2 Walsh spectra. The plotted Walsh tables show the same amplitude profile for PRESENT, GIFT, and SKINNY, which explains why purely amplitude-based diagnostics coincide for these examples.

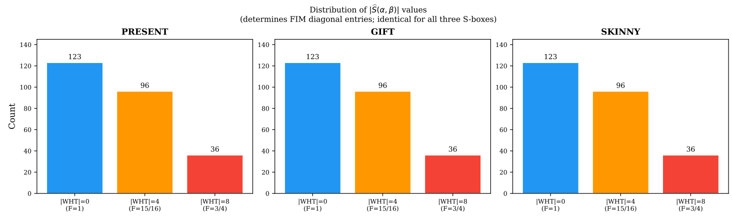

Fig. 3 Walsh amplitude distribution. The histogram records the shared counts of absolute Walsh coefficients over nonzero masks for the three tested S-boxes.

The three tested S-boxes have the same absolute Walsh-coefficient distribution over nonzero masks:

$$\#|\widehat{S}| = 0 : 123, \qquad \#|\widehat{S}| = 4 : 96, \qquad \#|\widehat{S}| = 8 : 36.$$

Therefore diagnostics depending only on the Walsh amplitudes coincide for these three examples. This shared spectral profile provides a useful baseline for reading the later CPA-specific quantities.

The boundary covariance matrix is built from Proposition 3.2. For $s = 4$ it is a 255×255 positive semidefinite matrix. Consistent with Theorem 6.1, all three matrices have rank 15 and active eigenvalues equal to 16 up to numerical precision.

Table 3 Boundary covariance structure for the tested 4-bit bijections.

S-box	Matrix size	Rank	Active eigenvalues
PRESENT	255×255	15	16 repeated 15 times
GIFT	255×255	15	16 repeated 15 times
SKINNY	255×255	15	16 repeated 15 times

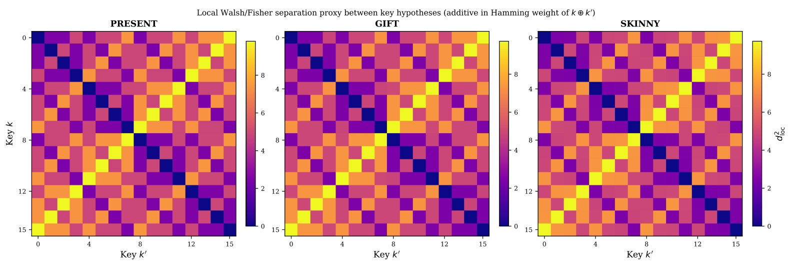

Fig. 4 Local key-separation proxy. The matrix visualises the covariance-weighted Walsh displacement between key hypotheses in the local Walsh/Fisher coordinates.

Table 4 Interpretation of the key-separation quantities.

Quantity	Status
$D_{\mathrm{KL}}(P_k \| P_{k'})$ for $k \neq k'$	Infinite, because supports are disjoint
$d_F(\delta_k, \delta_{k'})$ on the belief simplex	Exactly π
$d^2_{\mathrm{loc}}(k, k')$ in (19)	Local Walsh/Fisher covariance proxy

7.3 Local spectral separation

Using (19), we compute local diagonal Walsh/Fisher separation proxies between key hypotheses. This local covariance-weighted coordinate displacement is evaluated in the regularized Walsh/Fisher coordinates introduced earlier. The key-translation rule implies that the value depends only on the key difference $k \oplus k'$.

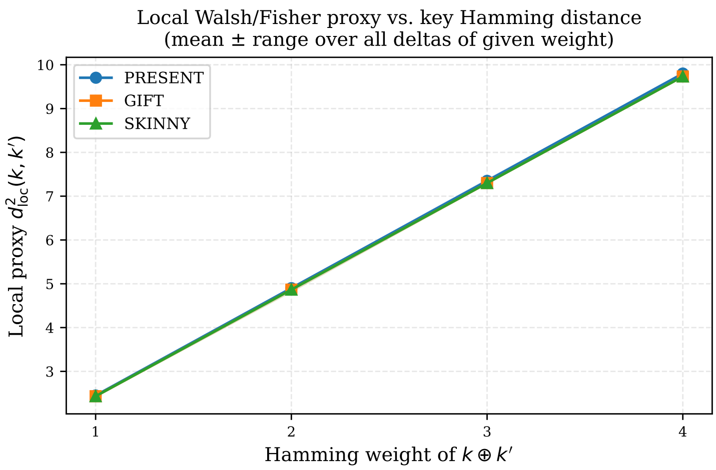

Fig. 5 Local proxy versus key-difference Hamming weight. The plot aggregates the same covariance-weighted Walsh displacement by the Hamming weight of $k \oplus k'$.

7.4 CPA leakage gaps

For bijective 4-bit S-boxes and uniform inputs, the scalar Hamming-weight variance in Proposition 5.1 is always

$$I_S(\mathrm{HW}, \sigma) = \frac{1}{\sigma^2}.$$

This scalar is a baseline leakage descriptor shared by PRESENT, GIFT, and SKINNY. Pairwise gaps

$$\Delta(\delta) = \frac{1}{16} \sum_{a \in \{0,1\}^4} \big(\mathrm{HW}(S(a)) - \mathrm{HW}(S(a \oplus \delta))\big)^2, \qquad \delta \neq 0,$$

are more informative for CPA because the expected log-Bayes gain against a wrong key with difference δ is $\Delta(\delta)/(2\sigma^2)$.

Table 5 shows how the scalar variance descriptor and the pairwise structure play complementary roles: all three S-boxes have the same average pairwise gap, while the worst and best key-difference cases vary. These pairwise gaps are the quantities that enter expected posterior-odds growth in the Gaussian leakage model.

Table 5 Pairwise Hamming-weight leakage gaps over nonzero key differences.

S-box	$\min_{\delta\neq 0}\Delta(\delta)$	$\max_{\delta\neq 0}\Delta(\delta)$	$\frac{1}{15}\sum_{\delta\neq 0}\Delta(\delta)$
PRESENT	1	7/2	32/15
GIFT	5/4	7/2	32/15
SKINNY	1	3	32/15

Table 6 Operational CPA interpretation of the pairwise gaps. The listed deltas are nonzero key differences $\delta = k \oplus k^*$ in hexadecimal.

S-box	Hardest gaps min Δ	Easiest gaps max Δ	Gap multiset over $\delta \neq 0$
PRESENT	1 at $\{A, E\}$	7/2 at $\{6\}$	1^2; $(5/4)^1$; $(3/2)^2$; $(7/4)^3$; $(5/2)^2$; $(11/4)^1$; 3^2; $(13/4)^1$; $(7/2)^1$
GIFT	5/4 at $\{C, E, F\}$	7/2 at $\{7\}$	$(5/4)^3$; $(3/2)^2$; 2^2; $(9/4)^3$; $(5/2)^2$; 3^2; $(7/2)^1$
SKINNY	1 at $\{5\}$	3 at $\{B\}$	1^1; $(5/4)^1$; $(3/2)^1$; $(7/4)^1$; 2^3; $(9/4)^2$; $(5/2)^3$; $(11/4)^2$; 3^1

Table 6 gives the CPA-oriented reading of the pairwise theory. PRESENT and SKINNY have a smaller hardest-key gap than GIFT under this leakage model, so the model-based estimate (34) assigns a larger trace budget to the hardest posterior-odds margin for those S-boxes. The comparison is local to the specified S-box computation and leakage model, which is the level at which CPA key hypotheses are evaluated.

7.5 Monte Carlo CPA validation

To connect the gap analysis to an attack observable, we simulate the Bayesian CPA model of Section 5. For each S-box, we use true key $k^* = 0$, uniformly sampled plaintext nibbles, leakage $W = \mathrm{HW}(S(a \oplus k^*)) + \varepsilon$, Gaussian noise standard deviation $\sigma = 2$, and 3000 Monte Carlo trials with seed 20260622. For every trace prefix, all 16 key log-likelihoods are accumulated and the true-key rank is recorded. The script `simulate_cpa.py` regenerates both Fig. 6 and the CSV summary `cpa_simulation_results.csv`.

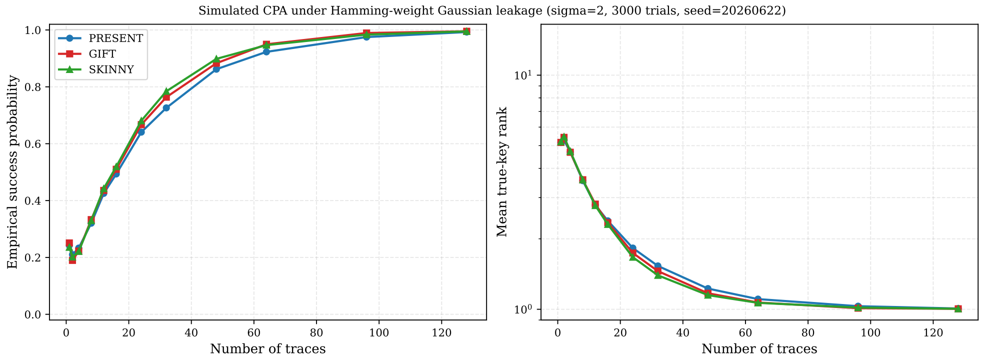

Fig. 6 Simulated CPA under Hamming-weight Gaussian leakage. The left panel shows empirical success probability and the right panel shows mean true-key rank over 3000 trials. The curves translate the pairwise-gap analysis into trace-dependent attack observables.

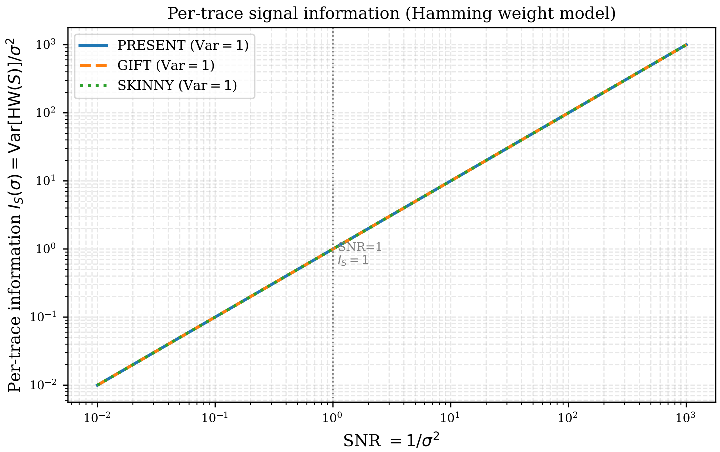

Fig. 7 Per-trace leakage baseline. The scalar Hamming-weight variance is the same for all bijective 4-bit S-boxes under uniform inputs, so CPA discrimination is completed by pairwise leakage gaps that determine key-hypothesis discrimination.

The simulation confirms the qualitative role of the pairwise gaps. At 64 traces, the empirical success probabilities are approximately 0.923 for PRESENT, 0.950 for GIFT, and 0.947 for SKINNY; at 128 traces, all three success probabilities are between 0.993 and 0.996. Mean true-key ranks decrease toward one for all three S-boxes. The finite-trace curves summarize the combined effect of all wrong-key competitors, while the minimum gap in Table 5 controls the hardest pairwise posterior-odds growth.

Table 7 Projected active-eigenspace curvature diagnostic for the tested 4-bit S-boxes.

S-box	Active dimension	Planes	Min K	Max K
PRESENT	15	105	-0.25	-0.25
GIFT	15	105	-0.25	-0.25
SKINNY	15	105	-0.25	-0.25

7.6 Active-eigenspace curvature

The curvature diagnostic of Section 6 is evaluated in the 15-dimensional active eigenspace of the boundary covariance matrix. The result is constant for all coordinate planes in the tested cases.

7.7 Reproducibility checklist

The computational claims in this section are reproducible from the additional files. The scripts `gen_figures.py`, `simulate_cpa.py`, `compute_curvatures.py`, and `verify_curvature.py` regenerate the figures, CPA simulation results, active covariance diagnostics, and curvature checks. The rank/eigenvalue statement is proved independently in Theorem 6.1; the scripts provide finite checks for the reported 4-bit examples.

8 Discussion

8.1 What the framework establishes

The framework establishes a controlled information-geometric model for the local S-box distributions that appear in side-channel key recovery. Its first role is structural: exact key-induced distributions are singular boundary points, while ordinary Fisher–Rao geometry belongs to full-support interiors or to the adversary's belief simplex. This distinction prevents assigning non-degenerate Fisher metrics, finite KL divergences, or ordinary geodesics to disjoint-support key endpoints.

The second role is algebraic. Walsh characters give explicit coordinates for the boundary orbit, and the key-translation rule determines how expectation coordinates move when the key hypothesis changes. This rule is elementary, but it is useful because it gives closed-form covariance and displacement expressions.

The third role is security-oriented. CPA is a Bayesian trajectory on the belief simplex. Under the Gaussian leakage model, pairwise Hamming-weight gaps determine expected log-Bayes gains against wrong keys. The scalar leakage variance is therefore a baseline descriptor, whereas pairwise gaps describe the discrimination structure relevant to key recovery.

This gives a concrete review criterion for the proposed quantities. A diagnostic is useful for side-channel security when it changes the interpretation of posterior-odds growth, hard key differences, or the validity of a geometric approximation. The numerical section combines the shared Walsh-amplitude invariants of PRESENT, GIFT, and SKINNY with the pairwise gap distributions and simulated CPA curves that determine which wrong keys are hardest to eliminate under the stated leakage model.

8.2 Position within S-box and side-channel analysis

The framework uses established 4-bit S-box classifications as background and focuses on the local distributions used in side-channel key recovery. PRESENT, GIFT, and SKINNY serve as reproducible lightweight examples with shared Walsh-amplitude profiles and different CPA gap distributions. The standard Gaussian Hamming-weight CPA model provides the inference setting in which the boundary S-box law, posterior belief law, and pairwise discrimination gaps are connected.

8.3 Cryptographic interpretation

For isolated bijective 4-bit S-boxes under uniform input and Hamming-weight leakage, $I_S(\mathrm{HW}, \sigma) = 1/\sigma^2$ for every bijection. This equality is best used as a baseline descriptor.

Practical side-channel behavior is shaped by pairwise leakage gaps, implementation leakage, noise, masking, alignment, higher-order effects, and the full cipher context.

In this sense, identical Walsh-amplitude histograms are a useful baseline. They show which quantities are shared by the three tested S-boxes, while the CPA gap distribution and Monte Carlo curves expose model-dependent hard and easy key differences. This is precisely the separation required before using information-geometric language in a cybersecurity setting.

The boundary covariance and local Walsh/Fisher separation proxy complement LAT, DDT, and empirical leakage evaluation. They provide coordinate diagnostics for the singular key orbit and inputs to posterior belief dynamics alongside standard cryptanalytic criteria.

8.4 Status of the curvature finding

The active-eigenspace value $K = -1/4$ is reported as a reproducible diagnostic under the projection, normalization, and sign conventions specified in Section 6. The rank and active eigenvalue statement is proved for every bijective S-box. A complete symbolic characterization of the curvature diagnostic, including the exact hypotheses under which it is constant, remains open.

8.5 Open problems

Several extensions are natural. First, the $\varepsilon \downarrow 0$ limit of regularized Fisher–Rao geometry could be studied using tools from singular information geometry. Second, the active-eigenspace curvature phenomenon could be attacked symbolically. Third, the leakage model could be extended to masked, higher-order, non-Gaussian, or profiled settings. Fourth, full-round ciphers can be handled by adding propagation through linear layers and key schedules.

9 Conclusion

We presented a boundary information-geometric framework for the local S-box distributions that arise in side-channel key recovery. The main modelling point is the separation between three spaces: the regular full-support Walsh exponential family, the singular key-induced boundary orbit, and the adversary's belief simplex. This separation makes Fisher, KL, and projection statements mathematically well scoped.

For bijective S-boxes, Walsh coordinates give closed-form boundary covariance matrices. The active covariance subspace has dimension $2^s - 1$ and all active eigenvalues equal 2^s in the unnormalised Walsh basis. Thus this part of the geometry is a general structural consequence of bijectivity.

For CPA, Bayesian updates occur on the belief simplex. Under Hamming-weight Gaussian leakage, the scalar variance information is a baseline fixed by bijectivity, while pairwise leakage gaps determine expected posterior-odds growth against wrong keys. Computations on PRESENT, GIFT, and SKINNY illustrate this distinction and reproduce the active curvature diagnostic $K = -1/4$ in the tested 4-bit cases.

The framework contributes a controlled geometric language for connecting singular S-box distributions, boundary covariance, and side-channel Bayesian inference. The security value of the framework is to make explicit which geometric quantities are structural invariants and which quantities affect local key-hypothesis discrimination under a specified leakage model.

Declarations

Availability of data and material

The datasets supporting the conclusions of this article are included within the article and its additional files. Additional file 1, `supplementary_reproducibility.pdf`, records the computational protocol, software environment, arithmetic conventions, and reproducibility coverage. Additional file 2, `supplementary_material.zip`, contains the scripts, dependency list, CPA simulation CSV, and reproduction README used to regenerate the numerical tables and figures.

Software information: project name, Boundary Information Geometry for S-box Side-Channel Leakage; project home page, https://anonymous.4open.science/r/IG-based-Sbox-SCL-DE49/; operating system, platform independent; programming language, Python 3; other requirements, `numpy`, `scipy`, and `matplotlib`; license, not specified; restrictions to use by non-academics, none beyond the repository license status.

Funding

Not applicable.

Acknowledgements

Not applicable.

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